

MSc Project Plan: Visual RAG - Autonomous Visual Search Agent for Video Retrieval

MSc Summer Project Plan

Autonomous Visual Search Agent for Video Retrieval (Visual RAG)

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Duration: 12 weeks (June – August 2026)

1. Project Overview

This project aims to build an autonomous Visual Retrieval-Augmented Generation (Visual RAG) agent that can intelligently search, retrieve, and reason over video content using multi-modal AI. Unlike traditional keyword-based video search, the system will combine visual understanding (CLIP, SigLIP), audio transcription, OCR, and large language models (LLMs) to enable natural language queries over video libraries with temporal precision.

The project bridges cutting-edge research in multi-modal retrieval, agentic AI, and video understanding — directly aligned with the supervisor's work at Memories.ai, a Series A startup building video intelligence platforms.

2. Research Questions

RQ1: How effectively can a Visual RAG pipeline retrieve relevant video segments from natural language queries compared to text-only retrieval baselines?

RQ2: What is the optimal embedding strategy for multi-modal video representation (visual + audio + text)?

RQ3: How can an agentic framework autonomously decompose complex queries into sub-retrieval tasks and synthesize coherent answers?

3. Technical Architecture

3.1 Video Ingestion Pipeline

- Frame extraction: Keyframe sampling using scene detection (PySceneDetect) and uniform temporal sampling
- Visual embeddings: CLIP ViT-L/14 or SigLIP for frame-level representations
- Audio transcription: Whisper (large-v3) for speech-to-text with timestamps
- OCR extraction: PaddleOCR or EasyOCR for on-screen text detection
- Metadata extraction: Duration, resolution, file info, chapter markers

3.2 Multi-Modal Embedding & Indexing

- Vector database: ChromaDB or Milvus for efficient similarity search with HNSW indexing
- Hybrid embeddings: Concatenated/fused visual + text + audio representations
- Temporal indexing: Segment-level embeddings with timestamp metadata for precise retrieval
- Sliding window approach: Overlapping temporal windows to capture cross-segment information

3.3 Visual RAG Agent

- Query understanding: LLM-based query decomposition (complex queries → atomic sub-queries)
- Multi-step retrieval: Iterative search refinement based on initial results
- Re-ranking: Cross-encoder re-ranking of candidate segments for precision
- Answer synthesis: LLM generates grounded answers with video segment citations and timestamps
- Tool use: Agent can invoke search, filter, compare, and summarize tools

3.4 User Interface

- Web-based demo: Streamlit or Gradio interface
- Natural language search bar with video playback at retrieved timestamps
- Confidence scores and source attribution for each answer

4. Project Timeline (12 Weeks)

Phase 1: Foundation (Weeks 1-3)

- Week 1: Literature review on Visual RAG, multi-modal retrieval, and video search systems. Set up development environment (GPU access, vector DB, Python env).
- Week 2: Implement video ingestion pipeline — frame extraction, Whisper transcription, OCR. Process test dataset (100+ videos).
- Week 3: Generate multi-modal embeddings (CLIP + text). Build vector database with temporal indexing. Baseline retrieval evaluation.

Phase 2: RAG Agent Development (Weeks 4-7)

- Week 4: Implement basic RAG pipeline — query → embed → retrieve → generate answer. Test with simple queries.
- Week 5: Add query decomposition module. Implement multi-step retrieval with iterative refinement.
- Week 6: Build cross-encoder re-ranking. Add temporal reasoning (e.g., "What happened after X?").
- Week 7: Integrate agentic framework — tool use, self-reflection, chain-of-thought retrieval. End-to-end testing.

Phase 3: Evaluation & Optimization (Weeks 8-10)

- Week 8: Create evaluation benchmark — 50+ queries across difficulty levels. Manual relevance annotations.
- Week 9: Quantitative evaluation: Recall@K, MRR, nDCG. Compare against text-only baselines. Ablation studies on modality contributions.
- Week 10: Optimize performance — embedding fusion strategies, retrieval speed, LLM latency. User study with 10+ participants.

Phase 4: Dissertation & Demo (Weeks 11-12)

- Week 11: Complete dissertation draft. Build polished demo with UI.
- Week 12: Final revisions, code cleanup, documentation. Submission.

5. Evaluation Metrics

Retrieval Quality:

- Recall@K (K=1, 5, 10): Proportion of relevant segments retrieved
- Mean Reciprocal Rank (MRR): Rank of first relevant result
- Normalized Discounted Cumulative Gain (nDCG): Graded relevance ranking quality

Answer Quality:

- Faithfulness: Is the generated answer grounded in retrieved content?
- Relevance: Does the answer address the query?
- Temporal Accuracy: Does the cited timestamp match the actual content?

System Performance:

- Query latency (end-to-end)
- Embedding generation throughput
- Index size and memory usage

6. Datasets

Primary: Custom video dataset curated from publicly available sources (lectures, tutorials, documentaries). Target: 200+ videos, 500+ hours.

Secondary: Existing benchmarks for comparison — ActivityNet-QA, MSRVTT-QA, or NExT-QA (if applicable).

7. Technology Stack

- Python 3.11+
- PyTorch (model inference)
- OpenAI CLIP / SigLIP (visual embeddings)
- Whisper large-v3 (ASR)
- ChromaDB / Milvus (vector store)
- LangChain or LlamaIndex (RAG framework)
- GPT-4o or Claude (LLM backbone for agent)
- Streamlit / Gradio (demo UI)
- Bristol HPC (BluePebble) for GPU compute

8. Key References

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[4] Radford, A. et al. (2023). Robust Speech Recognition via Large-Scale Weak Supervision. ICML. (Whisper)

[5] Bain, M. et al. (2021). Frozen in Time: A Joint Video and Image Encoder for End-to-End Retrieval. ICCV.

[6] Li, J. et al. (2023). BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. ICML.

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[9] Schick, T. et al. (2023). Toolformer: Language Models Can Teach Themselves to Use Tools. NeurIPS.

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9. Risks & Mitigations

Risk: GPU compute availability on BluePebble

Mitigation: Pre-compute embeddings early; use efficient models (CLIP ViT-B/32 as fallback)

Risk: LLM API costs for agent reasoning

Mitigation: Use open-source models (Llama 3, Mixtral) as alternatives; batch queries

Risk: Annotation effort for evaluation

Mitigation: Semi-automated annotation pipeline; smaller but high-quality benchmark

Risk: Scope creep with agent complexity

Mitigation: MVP-first approach — basic RAG first, then add agentic capabilities incrementally

10. Expected Contributions

1. A working Visual RAG system for natural language video retrieval with temporal precision

2. Empirical comparison of multi-modal embedding strategies for video search
3. An agentic framework for complex multi-step video queries
4. Open-source codebase and reproducible benchmark

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